Analytics & Big Data

Session 0: Course logistics

Prof. Dr. Gerit Wagner

(2026-03-23)

Welcome

Prof. Dr. Gerit Wagner

Academic background

- Universität Regensburg: Doctoral Student
- HEC Montréal: Postdoctoral Fellowship
- Otto-Friedrich-Universität Bamberg: Assistant Professor
- Frankfurt School of Finance & Management: Full Professor

Research interests

- Open, agentic, and boundary-spanning work
- AI-supported knowledge synthesis

Teaching interests

- Analytics and big data
- IT security
- Programming and software engineering
- Digital knowledge-intensive and platform-based work
- Literature review methods

About you

Please take a moment to think about the following questions. For each question, I'll invite a few of you to briefly share your thoughts.

1. Have you worked in an industry role or completed an internship with an analytics focus?

- What was something that felt **very useful** in your analytics work?
- Or where would you have liked to be **better prepared** for a data-related task?

2. Which analytics tools or programming languages have you used?

- What was a particularly interesting analytics **challenge** you encountered?
- Or a tool you found especially powerful or **surprising**?

3. What are your expectations for the course?

Analytics & Big Data

Analytics & Big Data in the curriculum

1*	Principles of Management	Financial Accounting	Introductionto Programming	Mathematics	Analytical Thinking & Critical Reasoning
2*	Principles of Finance	Microeconomy	Statistics& Probability	Macro-economics	Marketing
3*	Operations Management	Managerial Accounting	Markets, Incentives & Ethical Management	Applied Research Methods	BusinessLaw
4	Corporate Finance	Strategy & Organisation	Concentration		
5	SemesterAbroad &Internship Abroad				
6	Elective	Academic Skills Module	Concentration		
7	Elective	Elective	Elective	Thesis	

What exactly is "analytics"?

Business analytics refers to the computer-supported examination of data using mathematical models to drive decisions and actions within business situations (based on Davenport & Harris, 2007).

Related fields (based on Chen et al., 2012):

- **Business intelligence** refers to "the techniques, technologies, systems, practices, methodologies, and applications that analyze critical business data to help an enterprise better understand its business and market and make timely business decisions."
- **Data mining** refers to algorithmic techniques used to discover patterns, relationships, and knowledge from large datasets.
- **Data science** is the interdisciplinary field that combines statistical methods, data mining, machine learning, and database technologies to extract knowledge and insights from large and complex datasets.

Why analytics (and this course) matter

💰 Finance — Quantitative arbitrage at Princeton-Newport Partners

- Early pioneer of applying the Black-Scholes option pricing model in real markets
- Systematic identification of mispriced options through mathematical arbitrage
- Demonstrated how financial models and data could generate consistent trading profits

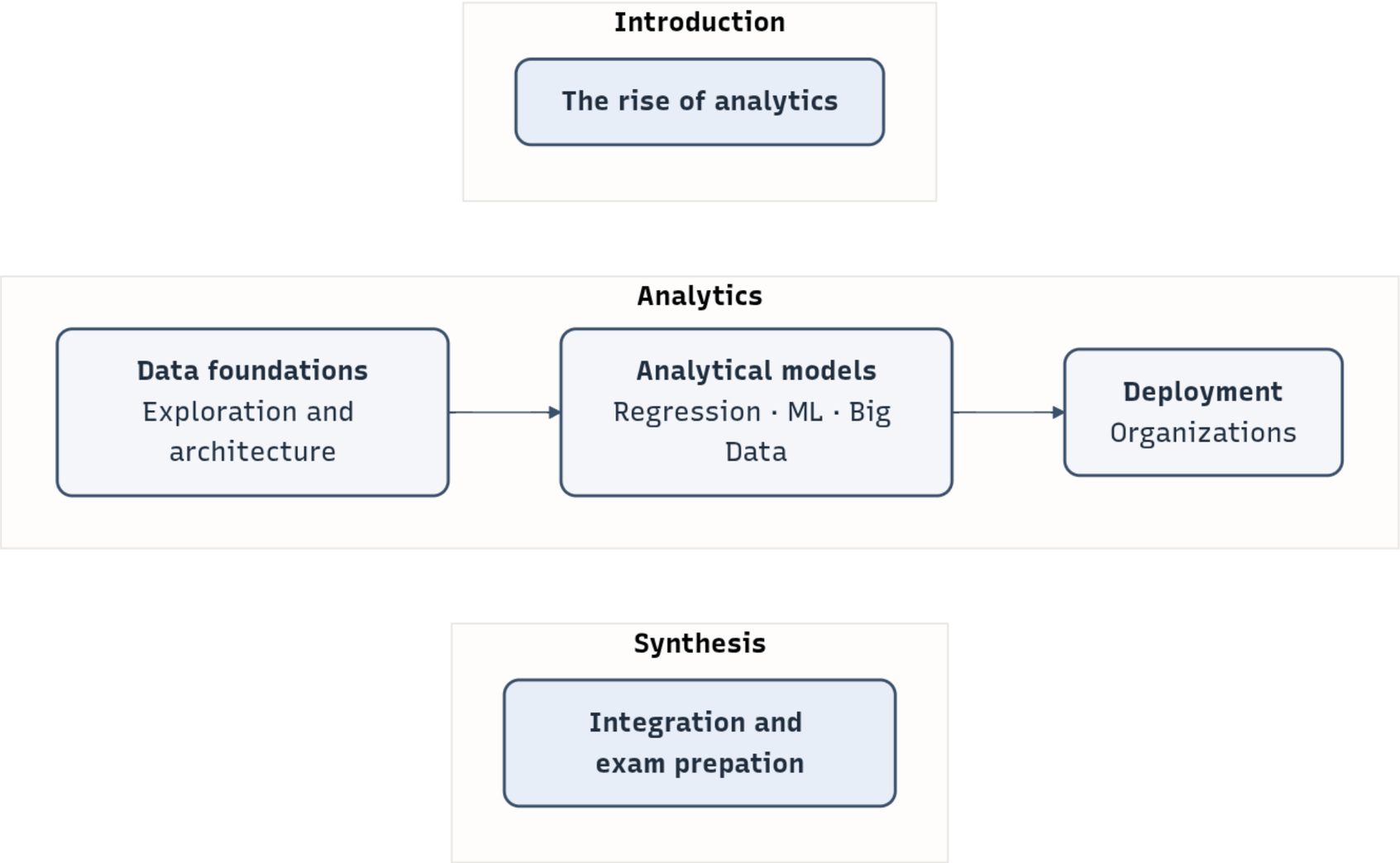
🎬 Entertainment — Personalization at scale at Netflix

- Recommendation systems shape what millions watch
- Continuous A/B testing and model refinement
- Data directly drives engagement and revenue

⚽ Sports — The Moneyball revolution at Oakland Athletics

- Advanced metrics replaced traditional scouting intuition
- Data-driven player selection with limited budget
- Competitive advantage through better measurement

Course architecture



What this course does *not* cover

- **Engineering of production-grade analytical systems**
  - Software engineering
  - Database design
  - DevOps/MLOps pipelines
  - Scalable application development
- **Statistical theory**
  - Inferential statistics
  - Proofs/derivations
- **Mathematical optimization and simulation models**
  - Scheduling, logistics, queueing
- **Autonomous or agentic decision systems**
  - Robots/agents acting on decisions
  - Reinforcement learning

💡 We will *touch* these areas where needed, but our focus is on **analytical reasoning for business decisions** and **notebook-based implementations**.

Course logistics

Course logistics

Workload:

- 150h total (44h in class, remaining: self-study and group project)

Assessment

- Exam: 60 minutes, 60 points (closed book)
- Group project: written paper (40 pages), 60 points (end of module)

Sessions:

- See overview in Canvas

Contact:

- EMail: g.wagner@fs.de
- Office hours: booking page

Individual circumstances

If you have family responsibilities, religious holidays, health-related matters, or other individual circumstances that may affect your participation or performance, please reach out early. We will work together to find a fair and workable solution.

Materials

Slides and materials

- Presentation slides and notebooks will be made available for download.

Short surveys at the end of each session

- If you notice anything that could benefit from further clarification or improvement, please take a note and let me know there.

Your input makes a real difference 🙏

Learning markers

- Facilitate learning and exam preparation.
- Help distinguish between illustrative material and key concepts and skills to prioritize.

> **Key concepts**
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> These markers highlight key skills and knowledge areas that you should prioritize when preparing for the exam.
> *Note*: This does not mean that other contents are excluded—they remain relevant for a complete understanding.

> **Learning focus**
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> These notes indicate how the content may be addressed in the exam and how you can prepare effectively.

Literature

- You are expected to take complementary notes and read the recommended literature.
- Literature and complementary materials will be listed at the end of each lecture.
- Reading of complementary materials depends on your interest and ambition.

Selected literature

- Part 1: Chen et al. (2012), Martínez-Plumed et al. (2019), Sharda et al. (2018)
- Part 2: Raschka (2015), Schutt & O'Neil (2014), Vaisman & Zimányi (2016)
- Part 3: Bengfort et al. (2018). Schmarzo (2016)
- Part 4: Davenport (2006), Vidgen et al. (2017)

You may also be interested in …

Bachelor's theses: See SuSy and additional information on this page.

In SuSy, you can find more information on my research topics:

- **Open, Agentic, and Boundary-Spanning Work**
  This area examines how AI agents and open digital infrastructures reshape knowledge work and collaboration across organizational boundaries. It focuses on agentic systems embedded in real work contexts (e.g., Git workflows, handbooks, and repositories) and how they transform coordination, governance, and accountability.
- **AI-Supported Knowledge Synthesis**
  This area investigates how AI supports knowledge synthesis across academic and professional contexts, focusing on transparency, rigor, and traceability in AI-assisted processes. It includes literature reviews and synthesis in digital gardens, second-brain systems, and other knowledge repositories.

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